

COMPLETE SOLUTION DOCUMENT

IndusMind

Unified Asset & Operations Intelligence Platform

AI-driven knowledge intelligence for asset-heavy industry

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Table of Contents

1. Introduction	3
1.1 Purpose	3
1.2 Background	3
1.3 Scope	3
2. Business Context	3
2.1 Problem Statement	3
2.2 Objectives and Success Criteria	3
2.3 Stakeholders and User Roles	4
3. Solution Overview	4
3.1 Guiding Design Principles	4
4. Functional Capabilities	5
4.1 Operational Walkthrough - Two Representative Journeys	5
5. Solution Architecture	6
5.1 Architecture Overview	6
5.2 Application Architecture	6
5.3 Information Architecture	7
5.4 AI Architecture	7
5.5 Integration Architecture	7
5.6 Security Architecture	7
5.7 Deployment Architecture	8
6. Implementation Detail	8
6.1 Implementation Approach	8
6.2 Presentation Tier	8
6.3 Application Tier	8
6.4 Data Tier	9
6.5 AI Subsystem	9
6.6 Service Interface Catalogue	9
6.7 Operations Engineering	10
6.8 Quality Assurance	10
6.9 Demonstration Scope	11
6.10 Implementation Blueprint - Recommended Construction Sequence	11
7. Non-Functional Requirements	12
8. AI Governance	13
9. Business Value	13
9.1 Illustrative Quantification	13
10. Delivery Roadmap	13
11. Risks and Mitigations	14
12. Assumptions, Dependencies and Constraints	14
13. Conclusion	14
Appendix A - Glossary	16
Appendix B - Demonstration Environment	16

1. Introduction

1.1 Purpose

This document describes the business context, functional scope, solution architecture, non-functional characteristics, governance model, and delivery roadmap of IndusMind, a unified asset and operations intelligence platform. It is the single authoritative description of the solution for evaluation purposes and is intended to be read alongside the working demonstration environment (Appendix B).

1.2 Background

Asset-heavy industries - refining, petrochemicals, power, manufacturing - operate on institutional knowledge distributed across technical documents, operational records, and the experience of senior personnel. Industry studies consistently indicate that technical professionals lose on the order of 35% of their working time locating information, that a typical plant operates seven to twelve disconnected repositories, and that an ageing expert workforce is creating a structural loss of tacit knowledge. IndusMind addresses this as a platform problem rather than a point-tool problem.

1.3 Scope

In scope	Out of scope (current release)
Ingestion and AI understanding of technical documents, including scanned legacy material	Physical document digitisation logistics (scanning services)
Conversational, citation-backed knowledge access for all plant roles	Autonomous control actions on plant systems
Maintenance operations: work orders, failure records, root-cause support, condition-based risk indication	Trained machine-learning failure models - a planned enhancement described in Section 10 (Delivery Roadmap); the current release provides transparent, rule-based risk indication
Regulatory compliance mapping, gap analysis, and audit evidence assembly	Legal interpretation of regulations
Operational knowledge capture: shift logbook, lessons learned	HR / workforce management functions
Enterprise platform services: multi-tenancy, RBAC, audit, configuration, integration APIs, reporting	Vendor-specific ERP adapters - generic API-key and webhook integration is provided today; certified adapters are a planned enhancement described in Section 10

2. Business Context

2.1 Problem Statement

Critical operating knowledge is fragmented, unreadable at scale, and eroding. The measurable consequences are extended mean-time-to-repair when reference material cannot be located, repetition of previously solved failures, protracted audit preparation, and the permanent loss of expertise at retirement. Existing document management and maintenance systems store information; none of them understand it or connect it.

2.2 Objectives and Success Criteria

#	Objective	Success criterion (measured in the demonstration environment)
O1	Reduce time-to-answer for technical questions	A cited, verifiable answer in under 60 seconds (baseline: hours of manual search)
O2	Make legacy documentation usable	A scanned manual becomes semantically searchable and

#	Objective	Success criterion (measured in the demonstration environment)
		citable within minutes of upload
O3	Provide early indication of equipment risk	Deteriorating condition trends raise an explained risk score and notification
O4	Reduce audit preparation effort	A complete evidence package assembled from live records in minutes
O5	Capture operational knowledge at source	A submitted shift log is citable by the assistant within minutes

2.3 Stakeholders and User Roles

The platform serves eight defined roles, each with a dedicated landing experience and a permission profile enforced at the interface and the API.

Role	Primary needs served
Plant Manager	Consolidated operational picture: health, risk, open work, compliance posture, cost of downtime
Maintenance Engineer	Investigation and planning: cited answers, failure history, root-cause support, risk indications
Field Technician	Execution on the plant floor: mobile work orders, meter readings, QR-based asset access, offline capability
Compliance Officer	Clause-level gap visibility, closure tracking, audit evidence assembly
Quality Engineer	Non-conformance management linked to assets, documents, and recurring patterns
Auditor	Read-only, traceable access to evidence and the audit trail
Operator	Digital shift logging and AI-summarised handover briefings
Administrator	Full runtime configuration of the platform without code changes

3. Solution Overview

IndusMind is a multi-tenant web platform that (i) ingests and understands an organisation's complete technical document base, (ii) links it with live operational data into a navigable knowledge graph, and (iii) exposes that knowledge through a conversational assistant whose every statement is grounded in a citable source. The same knowledge foundation powers operational capabilities: condition-based risk indication, root-cause analysis support, compliance gap detection, audit evidence assembly, and structured capture of day-to-day operating experience.

3.1 Guiding Design Principles

- **Evidence over assertion.** The assistant answers only from retrieved material; every claim carries a citation that is machine-verified before display, together with an explicit confidence indication.
- **Configuration over code.** Reference data, thresholds, formats, prompts, extraction rules, dashboard compositions, notification content, and schedules are data, administered at runtime. The platform contains no hardcoded business values.
- **Human accountability.** AI output informs decisions; it does not take them. Low-confidence extractions are routed to human review; analytical suggestions are advisory and attributed.
- **Enterprise controls first.** Tenant isolation, role-based authorisation, and an immutable audit trail are foundational services, not add-ons.

- **Environment parity.** One codebase operates identically on a single evaluation machine and on cloud infrastructure; only configuration differs.

4. Functional Capabilities

ID	Capability	Description
FC-01	Knowledge Assistant	Natural-language question answering across all documents and operational records; semantic and keyword retrieval fused, expanded through the knowledge graph; streamed answers with per-claim citations, confidence scoring, and user feedback capture.
FC-02	Document Ingestion	Managed intake of documents of any age: OCR for scans, layout and table parsing, semantic chunking, vectorisation, entity extraction using administrator-defined rules supplemented by AI, and graph linkage - with live progress and human review of low-confidence extractions.
FC-03	Knowledge Graph	A connected model of the plant - equipment, documents, failures, standards clauses, parts, lessons - explorable visually and used by the assistant for multi-step reasoning.
FC-04	Asset 360°	A complete per-asset view: hierarchy, documentation, work and failure history, condition trends against configured normal ranges, current risk indication, parts, and compliance items; printable QR labels resolve directly to this view from a mobile device.
FC-05	Maintenance Operations	Full work-order lifecycle with assignment, parts planning and consumption, and completion capture; structured failure recording; a root-cause workspace providing evidence-cited cause suggestions.
FC-06	Condition-Based Risk	Meter readings (entered, imported, or supplied via API) evaluated against configured ranges and failure history to produce explained risk scores and notifications.
FC-07	Compliance Management	Standards held clause-by-clause; automated mapping of clauses to plant evidence; side-by-side gap presentation in plain language; guided assembly of exportable audit evidence packages.
FC-08	Quality & Lessons	Non-conformance management and cross-incident pattern detection producing reusable, automatically surfaced lessons.
FC-09	Shift Logbook	Structured operator logs that are ingested into the knowledge base on submission - becoming citable within minutes - with AI-generated handover summaries.
FC-10	Dashboards & Search	Role-specific dashboards composed at runtime from a governed widget registry; global search across all entities; template-driven notifications with per-user channel preferences.
FC-11	Data Interchange & Reporting	Validated bulk import with column mapping and error reporting; export of any governed view; scheduled report generation and distribution.
FC-12	Administration	Runtime management of users, roles and permissions, reference data, settings hierarchy, AI model and prompt configuration, extraction rules, notification templates, translations, retention policies, API keys, and webhooks - through the interface, with full audit.

4.1 Operational Walkthrough - Two Representative Journeys

The journey of a document. A thirty-year-old scanned pump manual is uploaded. It is stored securely, read by optical character recognition, parsed for layout and tables, divided into meaning-preserving passages, vectorised for semantic retrieval, scanned for entities (asset tags, standard references, parameters) using administrator-defined rules supplemented by AI, and linked into the knowledge graph. Progress is visible at

every stage; extractions below the confidence threshold are queued for human confirmation. Within minutes the manual is searchable by meaning and citable by the assistant.

The journey of a question. An engineer asks: “Why does pump P-101 keep failing?” The platform authenticates the user, binds their tenant and role, retrieves relevant passages by combined semantic and keyword search, expands through the graph to the pump’s failure records and governing clauses, and generates an answer strictly from that evidence. The response streams back in seconds: a likely cause, the supporting history, the relevant manual section, and a recommended check - each statement carrying a verified citation, the whole answer carrying a confidence score. A task that previously consumed hours completes in under a minute.

5. Solution Architecture

5.1 Architecture Overview

The solution follows a layered architecture: client experiences, a presentation tier, a modular application tier, an asynchronous intelligence tier, and a polyglot data tier, with pluggable external AI providers and a uniform deployment model. Figure 1 presents the complete view; the amber sequence traces a single question end-to-end.

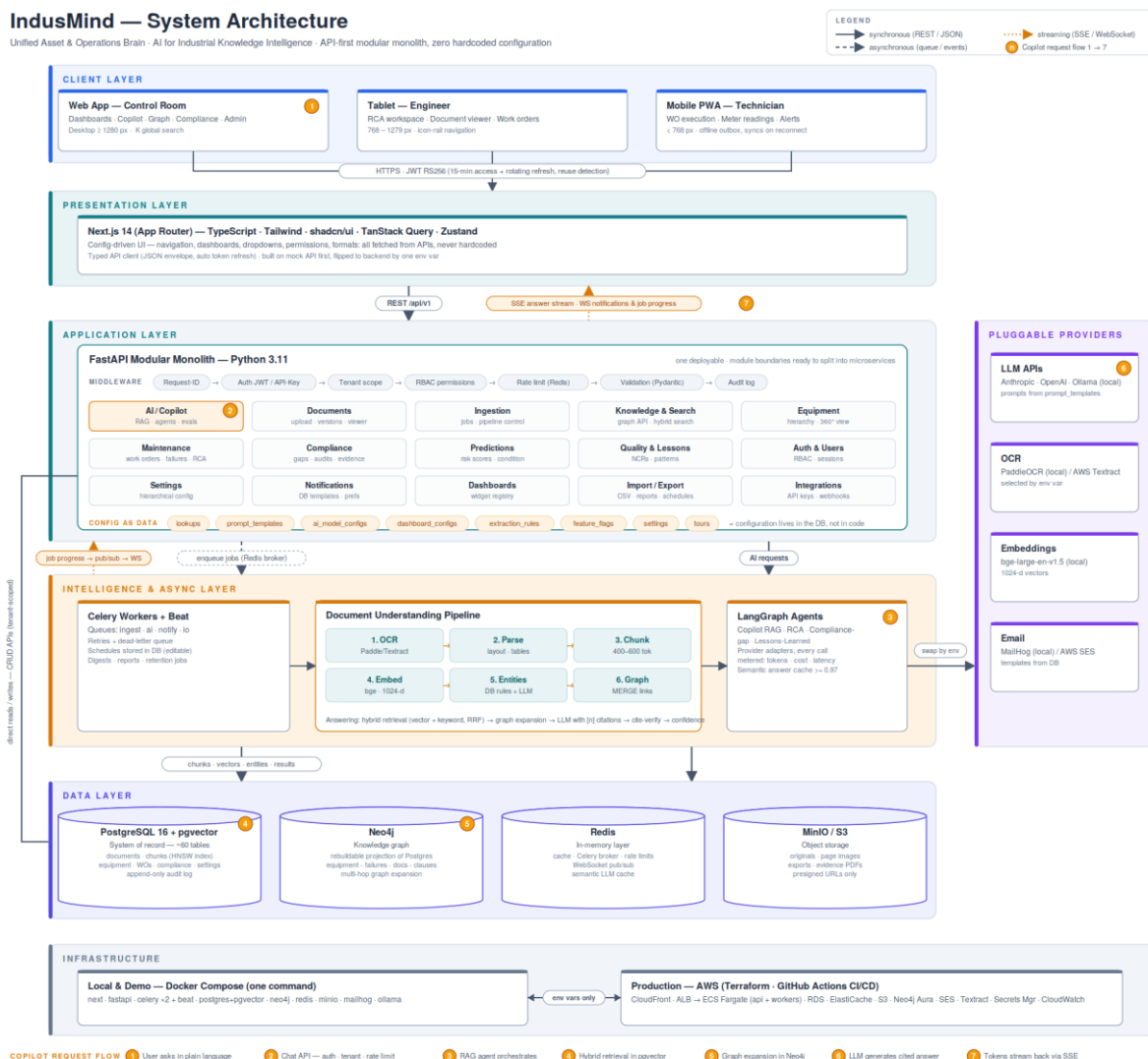


Figure 1 - IndusMind system architecture

5.2 Application Architecture

The application tier is a modular monolith: a single deployable composed of strictly bounded domain modules (knowledge and AI, documents and ingestion, equipment and condition, maintenance, compliance and quality,

operations, identity, and platform services), each owning its interface contracts, business logic, and data access. Module boundaries are enforced in code review and by an internal event bus for cross-module effects, preserving a low-friction path to extracting individual modules into services if scale demands. Every request traverses a uniform control chain - request identification, authentication, tenant binding, role authorisation, rate limiting, validation, and audit capture - before any business logic executes.

5.3 Information Architecture

Data domain	System of record	Notes
Master & transactional data (assets, work, compliance, users, configuration)	PostgreSQL 16	Approximately 60 relational entities; every record tenant-scoped; schema changes applied through versioned migrations
Semantic index	PostgreSQL (pgvector)	Passage embeddings stored beside their source records; approximate-nearest-neighbour indexed
Relationship model	Neo4j	A projection of the relational store, rebuildable at any time; used for graph exploration and retrieval expansion
Files & renditions	Object storage (S3-compatible)	Originals, page images, generated packages; access exclusively via expiring signed URLs
Volatile state	Redis	Caching, queues, rate counters, event fan-out

Within the relationship model, documents MENTION assets, parameters, and clauses (each link carrying confidence and page provenance); assets are PART_OF hierarchies; failure events OCCUR_ON assets and carry failure modes; work orders are PERFORMED_ON assets and may RESOLVE failure events; clauses GOVERN assets and procedures; and lessons are DERIVED_FROM the events and documents that produced them. These typed relationships allow the assistant to connect records that share no common wording.

5.4 AI Architecture

Document understanding is a six-stage asynchronous pipeline: optical character recognition, layout and table parsing, structure-aware chunking, embedding, entity extraction (administrator-defined rules first, AI second, human review for low confidence), and graph linkage. Question answering is retrieval-augmented: hybrid semantic-plus-keyword retrieval with rank fusion, graph-based expansion to related operational records, constrained generation from the retrieved evidence only, automated citation verification, confidence scoring, and token streaming to the client. Model access is mediated by a provider-adapter layer supporting hosted and locally executed models; prompts and model parameters are versioned configuration; identical or near-identical questions are served from a similarity-gated cache.

5.5 Integration Architecture

All functionality is exposed through versioned, documented REST APIs with a uniform response envelope. Machine-to-machine access uses scoped API keys; outbound integration uses signed webhooks with retry and delivery tracking, enabling enterprise asset-management systems to subscribe to platform events. Bulk interchange is provided through validated file import with interactive column mapping, governed exports, and scheduled reporting.

5.6 Security Architecture

Control area	Implementation
Identity & access	Short-lived signed access tokens with rotating refresh tokens and token-reuse detection; per-user session visibility and revocation; password policy and recovery

Control area	Implementation
	flows; fine-grained permission codes aggregated into the eight business roles and enforced at every endpoint
Tenant isolation	Tenant identity bound to the request context and applied structurally to all data access; isolation verified by dedicated automated tests
Data protection	Encryption in transit throughout and at rest in cloud deployment (managed keys); file access only via expiring signed URLs; secrets held in a managed vault; passwords hashed with a modern memory-hard algorithm; refresh tokens and integration keys stored only as hashes, sensitive configuration envelope-encrypted
Accountability	An append-only audit trail recording actor, action, subject, origin, and before/after state, available to authorised reviewers
AI-specific controls	Retrieved content handled strictly as data (prompt-injection resistance); model output schema-validated; per-user rate limits on AI endpoints; signed outbound webhooks

5.7 Deployment Architecture

Two deployment profiles run the identical build. The evaluation profile starts the complete platform - interface, services, workers, all data stores, object storage, and a local mail sink - on a single machine with one command, pre-populated with a representative model plant and role-based demonstration accounts. The production profile targets managed cloud services (container orchestration behind a load balancer, managed relational and cache services, object storage, managed graph database, transactional email) with infrastructure defined as code, an automated build-test-deploy pipeline, continuous backups, and health-probed, metric-instrumented operations. The profiles differ only in configuration values.

6. Implementation Detail

This section records how the architecture of Section 5 is realised, at the depth technical evaluators require, while remaining implementation-honest about demonstration scope (Section 6.9).

6.1 Implementation Approach

- **Contract-first.** A single versioned API contract - a uniform success envelope and a structured error envelope carrying machine-readable codes, field-level errors, and the request identifier - was fixed before construction. The interface and services were built independently against that contract and joined by configuration alone.
- **Configuration as data.** Reference lists, thresholds, formats, prompts, extraction rules, dashboard compositions, notification content, report definitions, and schedules are administered records, versioned where behaviourally significant. The codebase contains no business constants.
- **Bounded modules.** Each domain module owns its interface layer, service logic, data access, and schemas; cross-module effects travel over an internal event bus. The discipline is enforced in review and preserves an extraction path to services.

6.2 Presentation Tier

A modern TypeScript single-page application (server-rendered where beneficial) with a strict design-token system providing verified-contrast light and dark themes. On sign-in the client loads four runtime payloads - effective settings, the user's permission set, feature flags, and language bundles - and every formatted value, option list, dashboard, and visible control derives from them. Three form factors are deliberate: desktop control-room layouts, tablet layouts for engineering work, and an installable mobile experience for technicians with a queued offline outbox that reconciles on reconnect. Accessibility (WCAG 2.1 AA contrast, keyboard operability, reduced-motion support) is built into the component layer.

6.3 Application Tier

A Python 3.11 service exposing versioned REST with generated interface documentation. The uniform control chain of Section 5.2 is implemented as ordered middleware; business logic never executes before identity, tenancy, authorisation, limits, and validation have passed, and significant actions append to the audit trail with before/after state. Asynchronous work runs in dedicated worker pools over four queues (ingestion, AI, notification, interchange) with retry, backoff, and dead-letter handling; scheduled work - reports, digests, retention - executes from administered schedules. Worker progress is published to an in-memory event channel and fanned out to clients over WebSocket, which is how users observe ingestion live.

6.4 Data Tier

Domain group	Representative entities	Notes
Identity & access	tenants, users, roles, permissions, refresh-token families, reset tokens	Token families enable reuse detection; sessions are user-visible and revocable
Assets & condition	plants, equipment, meter definitions and readings	Normal operating ranges are configuration on the meter definition
Documents & knowledge	documents, versions, passages with embeddings, extracted entities, ingestion jobs	Passages retain page and position, enabling exact citation
Maintenance	work orders, parts and movements, failure records, analyses, risk indications	Work orders follow an enforced state machine; part consumption is transactional
Compliance & quality	standards, clauses, mappings, gaps, audits, evidence packages, non-conformances, lessons	Gap records bind clause to evidence with an explanation
Conversation & AI	sessions, messages, feedback, usage records, model configurations, prompt templates	Every model invocation is metered: tokens, computed cost, latency, cache use
Platform & configuration	settings hierarchy, reference lists, flags, dashboards, extraction rules, templates, views, translations, retention policies	The 'configuration as data' layer of Section 6.1
Integration & accountability	API keys, webhook endpoints and deliveries, import/export jobs, reports, audit log	The audit log is append-only

Approximately sixty relational entities, all tenant-scoped, evolved through versioned migrations. The graph store is populated exclusively by an idempotent projection job from the relational store and can be rebuilt at any time; object storage is addressed only through expiring signed URLs.

6.5 AI Subsystem

The ingestion pipeline and answering path of Section 5.4 are realised as queued worker stages with per-stage status, quality scoring on recognition output, and a human-review queue for low-confidence extractions. Retrieval fuses semantic and keyword result sets by reciprocal rank before graph expansion; generation is constrained to the retrieved evidence with an explicit instruction to abstain when it is insufficient; an automated verifier confirms each citation before display; a similarity-gated cache serves repeated questions. Prompts and model parameters are versioned configuration; the adapter layer meters every invocation and supports hosted providers and fully local execution interchangeably. Quality is scored continuously against a curated benchmark into which user feedback flows.

6.6 Service Interface Catalogue

All functionality is exposed as versioned REST with generated reference documentation; the principal service groups are:

Group	Representative operations
Identity & session	Sign-in, token refresh, sign-out, password recovery and change, session listing and revocation
Assistant & AI	Conversation sessions, streamed answers, answer feedback, administrative usage summaries
Documents & ingestion	Managed upload, document and version lifecycle, ingestion job control, extraction review
Assets & condition	Asset hierarchy and detail, code resolution and labels, meter definitions and readings
Maintenance	Work-order lifecycle and transitions, parts, failure records, analyses, risk indications
Compliance & quality	Standards and clauses, gap analysis, audits, evidence packages, non-conformances, lessons
Search & knowledge	Global search, graph queries, governed saved views
Platform services	Settings, reference data, dashboards and widget data, notifications and preferences, translations, imports, exports, reports
Integration & administration	API keys, webhook endpoints and deliveries, extraction rules, retention policies, audit-log access, bulk operations

6.7 Operations Engineering

Caching. Read-path caches (permission sets, reference data, dashboard data, asset hierarchies) carry purpose-specific lifetimes and are invalidated by domain events rather than timers alone. Assistant answers are additionally served from a similarity-gated cache: a question semantically near-identical to a recent one (within the same authorisation scope) replays the verified answer, materially reducing latency and model spend.

Rate governance. Sliding-window limits apply per user and route class - strictest on authentication and AI endpoints - with standards-compliant retry signalling, and each tenant carries a configurable daily model-token budget with a soft warning threshold and an administrative override.

Observability. Liveness and readiness probes verify every dependency; a metrics endpoint publishes request rate, latency, and error rate by route, queue depth, task duration by pipeline stage, model token consumption, and ingestion success rate; structured logs carry the request identifier end-to-end; production alerting triggers on error-rate, queue-depth, ingestion-failure, and latency thresholds, with centralised exception tracking.

Performance engineering. Asynchronous pooled data access; list endpoints use purpose-built queries with covering indexes and keyset pagination; vector search runs on tuned approximate-nearest-neighbour indexes with authorisation-scope pre-filtering; embeddings are computed in batches; smaller models are routed to classification and extraction with the larger model reserved for final answers; expensive dashboard aggregates are maintained as event-refreshed read models.

Delivery pipeline. Every change passes linting, the full test suite, and image build before deployment approval; images are minimal, non-root, and immutable, so any release can be redeployed exactly.

Backup and restoration. Continuous point-in-time database backup underwrites the 15-minute recovery-point objective; the rehearsed restoration sequence - restore snapshot, redeploy immutable images, rebuild the graph projection, run the verification suite - meets the 4-hour recovery-time objective, with cross-region restoration from replicated backups within 24 hours in the event of regional loss.

6.8 Quality Assurance

Layer	Coverage
Unit	Service and utility logic: settings resolution order, token rotation, chunking, cost arithmetic,

Layer	Coverage
	schedule parsing
Integration	Every endpoint's success and failure envelopes against a real database; the complete role-versus-endpoint authorisation matrix; tenant-isolation assertions
Pipeline	Golden-file verification: a reference document must yield expected passages, entities, and graph links; a submitted shift log must become citable
AI quality	The benchmark question set scored for answer quality, citation validity, and latency on every significant change
End-to-end	Scripted verification of the journeys in Section 4.1 and the objectives of Section 2.2

6.9 Demonstration Scope

The demonstration implements every capability in Section 4 end-to-end on a representative model plant. Choices made deliberately for this release, none requiring architectural rework: risk indication uses a transparent, explainable evaluation of trends and history (trained models are a planned enhancement, Section 10); federated single sign-on and multi-factor authentication are designed and surfaced but not active; interface translation is demonstrated in English and Hindi rather than a full catalogue; and enterprise asset-management connectivity is provided through the generic key-and-webhook layer rather than certified vendor adapters.

6.10 Implementation Blueprint - Recommended Construction Sequence

For a team recreating this solution, the following sequence minimises rework: each step yields a working, verifiable increment that the next step builds upon. Backend and interface proceed in parallel once the service contract of Section 6.1 is fixed.

Step	Build activity	Verified outcome
1	Provision the runtime: containers for the relational database (with the vector extension), cache, graph store, object storage, and a local mail sink; a single command starts everything	Empty platform starts cleanly on one machine
2	Scaffold the service: response envelope, error model, and the ordered control chain (request identity → authentication → tenant binding → authorisation → limits → validation → audit)	A health endpoint answers through the full chain
3	Identity and access: user accounts, token issuance and rotation, roles and permission codes, session management, password recovery; the append-only audit log	Sign-in works; every action is attributable
4	Configuration layer: hierarchical settings, reference lists, feature flags - and the administration screens that edit them	A value changed by an administrator changes behaviour everywhere, with no release
5	Asset registry and document intake: asset hierarchy, managed uploads via signed URLs, versioning, viewing	Documents and assets exist and are browsable
6	Ingestion pipeline, one stage at a time: recognition → parsing → chunking → embedding → entity extraction (rules, then AI, then human review) → graph linkage, with live progress to the client	An uploaded scan becomes searchable and graph-linked, watched in real time
7	Retrieval and the assistant: fused semantic-and-keyword search, graph expansion, evidence-constrained generation, citation verification, confidence scoring, streaming - and the benchmark that scores it	A plain-language question returns a cited, scored answer in seconds
8	Operations: work-order lifecycle, parts, failure records, root-cause workspace; meter definitions, readings, and rule-based	A deteriorating trend raises an explained, notified risk score

Step	Build activity	Verified outcome
	risk indication with notifications	
9	Assurance: standards and clauses, evidence mapping, gap analysis, evidence-package assembly; non-conformances and lessons	A gap is identified and an evidence package exported
10	Experience completion: role dashboards from the widget registry, global search, notification preferences, imports and exports, reports, shift logbook feeding the pipeline of step 6	Every role has a working, tailored experience
11	Hardening and cloud profile: rate governance, metrics and alerting, backup and rehearsed restoration, infrastructure as code, delivery pipeline	The identical build deploys to cloud by configuration only
12	Representative data and acceptance: seed a model plant (assets, documents, histories, condition trends, accounts per role) and execute the end-to-end verification of the journeys in Section 4.1	All success criteria of Section 2.2 pass on demand

The interface is built in a mirrored order: design tokens and the application shell first; the shared table, form, and chart components next; then the flagship screens in step order - document library and ingestion console, the assistant, asset views, operations, assurance - and finally role dashboards and administration. Building the interface against an internal simulation of the service contract, and switching to the live services by a single configuration value, allows the two streams to proceed independently and join without rework.

7. Non-Functional Requirements

ID	Category	Requirement	How the solution satisfies it
NFR-01	Performance	Interactive responses < 500 ms for standard operations; first token of an AI answer within seconds	Read-path caching; asynchronous execution of all heavy work; streamed generation; similarity-gated answer cache
NFR-02	Scalability	Linear scale-out with users and corpus size	Stateless application containers; per-queue worker pools; indexed vector search; read-replica-ready data tier; extractable modules
NFR-03	Availability	No single in-process dependency; graceful degradation of AI features	Queue-buffered processing; provider adapters permit failover between AI backends; health probes for orchestration
NFR-04	Security	Enterprise IAM, tenancy, and data protection (per Section 5.6)	Implemented as foundational middleware; verified by an authorisation test matrix and tenant-isolation tests
NFR-05	Auditability	Complete, tamper-evident record of user and administrative actions	Append-only audit log with before/after values; audit viewer for authorised roles
NFR-06	Maintainability	Behavioural change without code release	All business values externalised to administered configuration; versioned prompts, rules, and templates
NFR-07	Usability & accessibility	Role-appropriate experiences across desktop, tablet, and mobile; WCAG 2.1 AA contrast; light and dark themes	Responsive design system with verified contrast tokens; keyboard navigation; reduced-motion support
NFR-08	Offline tolerance	Field data capture must survive connectivity loss	Installable mobile experience with a queued outbox and reconciliation on reconnect
NFR-09	Portability	No binding to a single AI or infrastructure vendor	Provider adapters for models, OCR, and email; S3-compatible storage; infrastructure as code

ID	Category	Requirement	How the solution satisfies it
NFR-10	Data lifecycle	Governed retention and disposal	Administered retention policies per data class with archival to object storage and audited execution
NFR-11	Recoverability	RPO ≤ 15 minutes; RTO ≤ 4 hours (cloud profile)	Continuous backups; rebuildable graph projection; documented, rehearsed restoration procedure
NFR-12	Internationalisation	Interface language, units, and formats per user	Translation service and hierarchical settings resolved at sign-in; demonstrated in English and Hindi

8. AI Governance

The platform is designed for supervised AI adoption. Grounding is structural: the generation step receives only retrieved evidence and is instructed to abstain where evidence is insufficient; an automated verifier confirms that each citation exists and supports its statement before the answer is displayed. Confidence is disclosed on every answer. Human oversight is built into the workflows that matter: uncertain extractions require confirmation, and analytical outputs are advisory. Quality is measured continuously against a curated benchmark of reference questions, into which user feedback flows. Consumption is fully metered - every model invocation records tokens, computed cost, latency, and cache usage, giving administrators complete visibility of AI spend by feature, model, and day. Because prompts and model selections are versioned configuration, the organisation retains the freedom to change providers, adjust behaviour, or operate entirely on locally hosted models without code change.

9. Business Value

Value driver	Mechanism	Indicative effect
Professional productivity	Minutes-to-answer replaces hours-of-search	Recovers a substantial share of the ~35% of technical time currently lost
Availability & downtime	Early risk indication; immediate access to correct procedures and history	Reduced repeat failures; shorter mean-time-to-repair
Compliance economics	Continuous gap visibility; automated evidence assembly	Audit preparation reduced from weeks to hours; fewer findings
Knowledge continuity	Logs, analyses, and lessons captured in citable form	Expertise retained beyond individual tenure
Maintenance efficiency	History-informed planning; parts consumption visibility	Reduced emergency work; improved stock decisions

9.1 Illustrative Quantification

For a single plant employing 50 engineers at a loaded annual cost of ₹25 lakh each, with 35% of time lost to information search: recovering a conservative half of that loss yields approximately ₹2.2 crore per annum in productivity value alone - before attributing any avoided downtime, where a single prevented day in a process plant commonly exceeds the platform's annual cost.

10. Delivery Roadmap

Phase	Scope	Exit criteria
1 - Foundation	Identity, tenancy, RBAC, audit; asset registry; document intake and viewing	Secure, populated system of record

Phase	Scope	Exit criteria
2 - Intelligence	Ingestion pipeline; semantic search; knowledge graph; cited assistant	Objectives O1 and O2 demonstrably met
3 - Operations	Work management, failure records, root-cause support, condition data and risk indication	Objective O3 demonstrably met
4 - Assurance	Compliance mapping, gap analysis, evidence packages; quality and lessons	Objective O4 demonstrably met
5 - Enterprise readiness	Shift logbook; interchange and reporting; integrations; internationalisation; hardening	Objective O5 met; production profile validated

The demonstration environment implements all five phases end-to-end at evaluation depth. Beyond the current release, the next planned enhancements are: trained predictive models built on the condition data the platform accumulates in operation; federated single sign-on and multi-factor authentication (already designed and surfaced in the product today); certified adapters for enterprise asset-management systems; and expanded interface-language coverage.

11. Risks and Mitigations

ID	Risk	L	I	Mitigation
RK-01	AI answer inaccuracy erodes user trust	M	H	Structural grounding; automated citation verification; visible confidence; abstention on weak evidence; continuous benchmark measurement with user-feedback intake
RK-02	Poor OCR quality on degraded legacy scans	M	M	Layout-aware processing; per-page quality scoring; human review queue; re-processing supported as rules and engines improve
RK-03	AI operating cost escalation	M	M	Full per-call metering with administrative visibility; similarity-gated caching; per-user rate limits; local-model option
RK-04	Data sensitivity and confidentiality concerns	L	H	Tenant isolation by construction; role-based access; complete audit trail; option to operate all AI components on-premises
RK-05	Adoption resistance from field personnel	M	M	Role-tailored, mobile-first experiences; QR-based zero-navigation access; offline tolerance; guided onboarding
RK-06	Vendor dependency (AI or cloud)	L	M	Provider-adapter layer; S3-compatible storage; infrastructure as code; demonstrated fully-local operation

L = likelihood, I = impact (H/M/L).

12. Assumptions, Dependencies and Constraints

- Source documents are made available in electronic form (native or scanned); physical scanning is outside scope.
- Standards and regulatory texts used for compliance mapping are licensed to the operating organisation.
- Condition-based risk indication depends on the availability of meter data (manual entry, import, or API).
- Hosted AI models require outbound connectivity; a fully local model profile is available where connectivity or data-residency constraints apply.
- Named user counts, retention periods, and notification policies are configured per organisation during onboarding.

13. Conclusion

IndusMind converts an eroding liability - fragmented, unreadable, departing knowledge - into a governed operating asset: one platform in which every document is understood, every answer is evidenced, operational risk is watched continuously, and daily experience is retained. The solution is architected to enterprise expectations - isolated, authorised, audited, configurable, portable - and every objective in Section 2.2 is verifiable in the accompanying demonstration environment within minutes.

Appendix A - Glossary

Term	Definition
OCR	Optical character recognition - extraction of machine-readable text from scanned images
Semantic search	Retrieval by meaning using vector representations, rather than by exact keywords
Retrieval-augmented generation	An AI pattern in which the model answers strictly from retrieved source passages, enabling citation
Knowledge graph	A stored network of entities (assets, documents, failures, clauses) and their relationships
Entity extraction	Automated identification of meaningful items - asset tags, clause references, parameters - within text
Confidence score	The system's disclosed estimate of how well an answer is supported by evidence
Multi-tenancy	One installation serving multiple organisations with strict logical data separation
RBAC	Role-based access control - permissions granted through defined roles
Webhook	A signed, machine-to-machine notification emitted by the platform on defined events
RPO / RTO	Recovery point objective (maximum data loss) and recovery time objective (maximum restoration time)

Appendix B - Demonstration Environment

The evaluation build starts with a single command and is pre-populated with a representative model plant: an asset hierarchy, twelve technical documents of varying age and quality, three years of maintenance and failure history, ninety days of condition data exhibiting a deteriorating trend, an open compliance clause, and a recent shift log. Demonstration accounts are provided for the administrator, plant manager, maintenance engineer, field technician, and compliance officer roles, allowing every objective and capability described in this document to be verified directly.